

**SAFETY DATA SHEET**

Revision Date 01-Apr-2024

Revision Number 6

**1. Identification**

**Product Name** Chloroacetaldehyde, 45% Aqueous Solution

**Cat No. :** A11037

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**2. Hazard(s) identification****Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                                  |              |
|--------------------------------------------------|--------------|
| Flammable liquids                                | Category 4   |
| Acute oral toxicity                              | Category 3   |
| Acute dermal toxicity                            | Category 3   |
| Acute Inhalation Toxicity - Vapors               | Category 2   |
| Skin Corrosion/Irritation                        | Category 1 B |
| Serious Eye Damage/Eye Irritation                | Category 1   |
| Carcinogenicity                                  | Category 2   |
| Specific target organ toxicity (single exposure) | Category 3   |
| Target Organs - Respiratory system.              |              |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Combustible liquid  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
Fatal if inhaled  
Suspected of causing cancer  
Toxic if swallowed or in contact with skin

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Do not breathe dust/fume/gas/mist/vapors/spray  
Use only outdoors or in a well-ventilated area  
Wear respiratory protection  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep cool

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

Rinse mouth  
Do NOT induce vomiting

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life

### 3. Composition/Information on Ingredients

| Component          | CAS No    | Weight % |
|--------------------|-----------|----------|
| Water              | 7732-18-5 | 55.00    |
| Chloroacetaldehyde | 107-20-0  | 45.00    |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                             |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                    |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                   |
| <b>Inhalation</b>                          | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.                                                                                |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                         |

## 5. Fire-fighting measures

|                                         |                                                                                                                     |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                            |
| <b>Flash Point</b>                      | 62 °C / 143.6 °F                                                                                                    |
| <b>Method -</b>                         | No information available                                                                                            |
| <b>Autoignition Temperature</b>         | 620 °C / 1148 °F                                                                                                    |
| <b>Explosion Limits</b>                 |                                                                                                                     |
| <b>Upper</b>                            | No data available                                                                                                   |
| <b>Lower</b>                            | No data available                                                                                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                            |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                            |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Combustible material. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Hydrogen chloride. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 4             | 2                   | 0                  | -                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.          |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.                                                                                                                             |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Storage.</b> | Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Strong bases. Strong oxidizing agents. Metals.                                                                                                       |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component          | ACGIH TLV      | OSHA PEL                                                                                                             | NIOSH                                                          | Mexico OEL (TWA) |
|--------------------|----------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|------------------|
| Chloroacetaldehyde | Ceiling: 1 ppm | Ceiling: 1 ppm<br>Ceiling: 3 mg/m <sup>3</sup><br>(Vacated) Ceiling: 1 ppm<br>(Vacated) Ceiling: 3 mg/m <sup>3</sup> | IDLH: 45 ppm<br>Ceiling: 1 ppm<br>Ceiling: 3 mg/m <sup>3</sup> | Ceiling: 1 ppm   |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

|                             |                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Engineering Measures</b> | Ensure that eyewash stations and safety showers are close to the workstation location.<br>Ensure adequate ventilation, especially in confined areas. |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|

### Personal Protective Equipment

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face Protection</b>      | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                       |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Recommended Filter type:</b> | Particulates filter conforming to EN 143.                                                                                                                                                                                                         |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                       |        |
|-----------------------|--------|
| <b>Physical State</b> | Liquid |
|-----------------------|--------|

|                                        |                            |
|----------------------------------------|----------------------------|
| Appearance                             | Colorless                  |
| Odor                                   | pungent                    |
| Odor Threshold                         | No information available   |
| pH                                     | 1.2 (10 g/l @ 20°C)        |
| Melting Point/Range                    | -28 °C / -18.4 °F          |
| Boiling Point/Range                    | 80 - 100 °C / 176 - 212 °F |
| Flash Point                            | 62 °C / 143.6 °F           |
| Evaporation Rate                       | No information available   |
| Flammability (solid,gas)               | Not applicable             |
| Flammability or explosive limits       |                            |
| Upper                                  | No data available          |
| Lower                                  | No data available          |
| Vapor Pressure                         | 23 hPa @ 20 °C             |
| Vapor Density                          | No information available   |
| Specific Gravity                       | 1.214 g/cm3                |
| Solubility                             | No information available   |
| Partition coefficient; n-octanol/water | No data available          |
| Autoignition Temperature               | 620 °C / 1148 °F           |
| Decomposition Temperature              | No information available   |
| Viscosity                              | No information available   |

## 10. Stability and reactivity

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                 |
| Stability                        | Stable under normal conditions.                                            |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition.          |
| Incompatible Materials           | Strong bases, Strong oxidizing agents, Metals                              |
| Hazardous Decomposition Products | Hydrogen chloride, Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                   |
| Hazardous Reactions              | None under normal processing.                                              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                     |
|-------------|-------------------------------------|
| Oral LD50   | Category 3. ATE = 50 - 300 mg/kg.   |
| Dermal LD50 | Category 3. ATE = 200 - 1000 mg/kg. |
| Vapor LC50  | Category 2. ATE = 0.5 - 2 mg/l.     |

#### Component Information

| Component          | LD50 Oral               | LD50 Dermal                | LC50 Inhalation                              |
|--------------------|-------------------------|----------------------------|----------------------------------------------|
| Water              | -                       | -                          | -                                            |
| Chloroacetaldehyde | LD50 = 23 mg/kg ( Rat ) | LD50 = 67 mg/kg ( Rabbit ) | LC50 650 - 780 mg/m <sup>3</sup> ( Rat ) 1 h |

|                                      |                          |
|--------------------------------------|--------------------------|
| Toxicologically Synergistic Products | No information available |
|--------------------------------------|--------------------------|

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component | CAS No | IARC | NTP | ACGIH | OSHA | Mexico |
|-----------|--------|------|-----|-------|------|--------|
|-----------|--------|------|-----|-------|------|--------|

|                    |           |            |            |            |            |            |
|--------------------|-----------|------------|------------|------------|------------|------------|
| Water              | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Chloroacetaldehyde | 107-20-0  | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component          | Freshwater Algae | Freshwater Fish                             | Microtox   | Water Flea |
|--------------------|------------------|---------------------------------------------|------------|------------|
| Chloroacetaldehyde | Not listed       | LC50: = 42.7 mg/L, 96h static (Danio rerio) | Not listed | Not listed |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

| Component          | log Pow |
|--------------------|---------|
| Chloroacetaldehyde | 0.62    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2232  
 Proper Shipping Name 2-CHLOROETHANAL SOLUTION  
 Hazard Class 6.1  
 Packing Group I

### TDG

Forbidden

### IATA

FORBIDDEN FOR IATA TRANSPORT

UN-No UN2232  
 Proper Shipping Name 2-CHLOROETHANAL SOLUTION FORBIDDEN FOR IATA TRANSPORT  
 Hazard Class 6.1  
 Packing Group I

### IMDG/IMO

**UN-No** UN2232  
**Proper Shipping Name** 2-CHLOROETHANAL SOLUTION  
**Hazard Class** 6.1  
**Packing Group** I

## 15. Regulatory information

### United States of America Inventory

| Component          | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|--------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Water              | 7732-18-5 | X    | ACTIVE                                        | -                           |
| Chloroacetaldehyde | 107-20-0  | X    | ACTIVE                                        | -                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

### International Inventories

X = listed.

| Component          | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|--------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Water              | 7732-18-5 | X   | -    | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Chloroacetaldehyde | 107-20-0  | -   | X    | 203-472-8 | X     | X    | X    | X    | X     | KE-05490 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### SARA 311/312 Hazard Categories

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

Not applicable

**Clean Air Act**

Not applicable

**OSHA** - Occupational Safety and Health Administration

Not applicable

#### CERCLA

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component          | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|--------------------|--------------------------|-------------------------------------------|-------------------------------|
| Chloroacetaldehyde | 1000 lb                  | -                                         | 1000 lb                       |

|  |  |  |        |
|--|--|--|--------|
|  |  |  | 454 kg |
|--|--|--|--------|

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component          | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|--------------------|---------------|------------|--------------|----------|--------------|
| Water              | -             | -          | X            | -        | -            |
| Chloroacetaldehyde | X             | X          | X            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH**

| Component          | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water              | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Chloroacetaldehyde | 107-20-0  | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component          | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Water              | 7732-18-5 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Chloroacetaldehyde | 107-20-0  | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water     | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |



|                    |          |                |                |                |                |
|--------------------|----------|----------------|----------------|----------------|----------------|
| Chloroacetaldehyde | 107-20-0 | Not applicable | Not applicable | Not applicable | Not applicable |
|--------------------|----------|----------------|----------------|----------------|----------------|

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Revision Date** 01-Apr-2024  
**Print Date** 01-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**